The Origin of PhiBase

Building an Exact Arithmetic One Step at a Time

James A. Hinds

Contents

Stepping Stone 1

Every Length Needs a Name

The journey that led to PhiBase did not begin with numbers. It began with two triangles.

Imagine holding a golden triangle in your hand. Not a drawing, but a real one built from three straight pieces. Two of those pieces are the same length, while the third is shorter. Now make another triangle just like it. Join them together. Make another, and keep building.

Very quickly, something remarkable happens. No matter how many triangles you assemble, every straight edge in the construction is made from only two lengths. There are short pieces and long pieces, and no third length is needed.

Let the short length be

$$1,$$

and let the long length be

$$\varphi = \frac{1+\sqrt{5}}{2}.$$

A straight run containing seven short pieces and four long pieces has total length

$$7 + 4\varphi.$$

PhiBase records those two totals as

$$P(7,4).$$

The first coordinate counts the total number of short units embedded in the run. The second counts the total number of long units. The notation does not record the order in which the pieces appear. Many different arrangements can have the same PhiBase value.

For example, the sequences

$$1,\ \varphi,\ \varphi,\ 1,\ 1$$

and

$$\varphi,\ 1,\ 1,\ 1,\ \varphi$$

both contain three short units and two long units. Both therefore have the same PhiBase value,

$$P(3,2).$$

They are different paths, but they have the same total embedded length.

Figure 1: A straight sequence of short and long steps. Each marked point shows the accumulated PhiBase value of all steps before it.

The marked points in Figure 1 are cumulative totals. Each new short step adds

$$P(1,0),$$

and each new long step adds

$$P(0,1).$$

Thus

$$P(0,0) = 0$$

means no length at all,

$$P(1,0) = 1$$

means one short unit, and

$$P(0,1) = \varphi$$

means one long unit.

Nothing new has been invented. The value $P(n,p)$ is simply the exact embedding

$$n + p\varphi.$$

Its usefulness is that the two kinds of length remain visible instead of being blurred into a decimal approximation.

Builder's Notebook

Write the PhiBase value of each collection:

two short units,

one short unit and one long unit,

five long units,

four short units and three long units,

and

no units at all.

Then write two different arrangements that both have the value

$$P(4,3).$$

Looking Ahead

PhiBase can name the total length of any straight collection of short and long units. The next question is what happens when two such totals are joined.

Stepping Stone 2

What Happens When We Join Two Collections?

In the previous stepping stone we learned that a PhiBase quantity records an embedding. The pair

$$P(n, p)$$

does not describe the order of the pieces. It records only the total number of short units and the total number of long units contained in the construction.

That observation makes addition almost unavoidable.

Suppose one collection contains

$$P(3, 1),$$

and a second contains

$$P(2, 4).$$

Imagine placing all of the pieces on the same workbench.

Nothing has changed.

The short pieces are still short.

The long pieces are still long.

The only question is how many of each kind are now present.

There are

$$3 + 2 = 5$$

short units and

$$1 + 4 = 5$$

long units.

The combined collection is therefore

$$P(5, 5).$$

Notice what did *not* matter.

We never asked which piece came first.

We never asked which piece came last.

Order belongs to a path.

Addition belongs to a collection.

The embedding simply counts the total amount of each kind of length.

The same construction works for every pair of PhiBase quantities.

If one collection is

$$P(a, b)$$

and the other is

$$P(c, d),$$

their combined embedding is

$$P(a, b) + P(c, d) = P(a + c,\ b + d).$$

Nothing mysterious has happened.

The short counts combine with the short counts.

The long counts combine with the long counts.

That is exactly what the construction itself tells us to do.

Subtraction follows the same reasoning.

If a collection described by

$$P(8,7)$$

contains another collection described by

$$P(1,5),$$

removing those pieces leaves

$$P(8,7) - P(1,5) = P(7,2).$$

Once again, only the totals change.

The embedding remembers how much material remains.

It says nothing about the order in which the pieces were originally assembled.

That distinction becomes important later.

When we begin walking through a construction, the order of the steps creates an address.

Today we are not walking.

We are only counting.

Builder's Notebook

Combine the following collections.

$$P(2,3) + P(5,1),$$

$$P(8,0) + P(0,4),$$

$$P(6,5) - P(2,1),$$

$$P(9,7) - P(4,6).$$

Now write two different arrangements of short and long pieces that both represent

$$P(4,3).$$

The arrangements are different.

The embedding is the same.

Looking Ahead

Adding collections changes only the totals.

The next question is very different.

What happens when every short piece grows into a long one, and every long piece grows by the same amount?

Stepping Stone 3

What Happens When Everything Grows?

Until now nothing has changed except the number of pieces.

Addition combined two collections.

Subtraction removed one collection from another.

The pieces themselves never changed.

Growth is different.

Suppose every short edge grows until it becomes exactly one long edge.

That is easy to picture.

Now ask the same question about a long edge.

What does it become?

Do not guess.

Look at Figure 2.

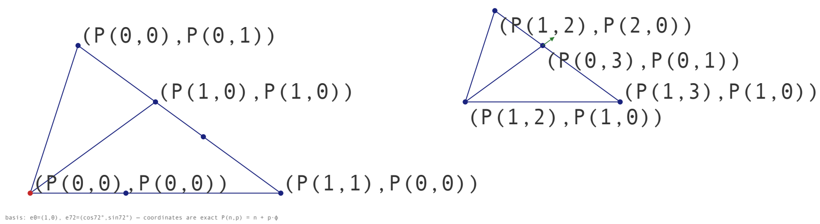

Figure 2: Growing a golden triangle. The enlarged construction is built from the same two edge lengths.

The picture answers both questions.

A short edge becomes one long edge.

A long edge becomes one short edge and one long edge.

Nothing else appears.

No third length is created.

The same two pieces simply rearrange themselves.

The familiar identity

$$\varphi^2 = \varphi + 1$$

is nothing more than a compact description of that picture.

The geometry came first.

The equation came afterward.

Growing One Collection

Suppose a collection contains

$$P(3,2).$$

The three short pieces each become one long piece.

The two long pieces each become one short piece and one long piece.

The result is easily counted.

Before	$\longrightarrow$	After
3 short		3 long
2 long		2 short + 2 long
		2 short + 5 long

The grown collection is therefore

$$P(2,5).$$

Nothing has been multiplied.

Nothing has been approximated.

We simply counted what the geometry produced.

Exactly the same reasoning works for every PhiBase collection:

$$P(n,p) \longrightarrow P(p,n+p).$$

The rule is only a summary.

The construction is the real mathematics.

Following the Growth

Beginning with one short unit,

$$P(1,0),$$

the first few growth steps are

Before	After
$P(1,0)$	$P(0,1)$
$P(0,1)$	$P(1,1)$
$P(1,1)$	$P(1,2)$
$P(1,2)$	$P(2,3)$
$P(2,3)$	$P(3,5)$

Every row is produced by exactly the same construction.

The arithmetic merely keeps count.

Builder's Notebook

Continue the table six more rows.

Do not use decimal numbers.

Do not multiply.

Simply let every short edge become one long edge, and every long edge become one short edge and one long edge.

The geometry will produce the arithmetic for you.

Looking Ahead

Growing by φ turned out to be surprisingly gentle.

Was that simply good fortune?

Or can every PhiBase multiplication be understood in the same way?

Stepping Stone 4

Can Every Multiplication Behave Like Growth?

Growing by φ turned out to be wonderfully simple. Every short piece became one long piece, and every long piece became one short piece and one long piece. The geometry performed the work. PhiBase merely counted the result.

That success invites a new question.

Was growing special because one of the quantities happened to be φ, or can every multiplication be understood in the same spirit?

Suppose we wish to multiply

$$P(a, b)$$

by

$$P(c, d).$$

Instead of trying to understand the whole product at once, let us pretend we are still growing. During growth, one quantity had only a single nonzero component. Can we arrange for the same thing to happen here?

That is the real problem.

We are looking for a companion that allows one part of the multiplication to disappear, leaving something as simple as growth.

For the moment, however, let us simply build the product.

Write the two quantities as

$$P(a,b) = a + b\varphi,$$

and

$$P(c,d) = c + d\varphi.$$

Now expand them exactly as you would any two binomials,

$$(a + b\varphi)(c + d\varphi) = ac + ad\varphi + bc\varphi + bd\varphi^2.$$

Only one unfamiliar term appears,

$$\varphi^2.$$

The previous stepping stone has already shown us what to do.

Replace it with

$$\varphi + 1.$$

Now collect the short pieces together, then collect the long pieces.

$$\begin{aligned} \text{Short part} &= ac + bd \\ \text{Long part} &= ad + bc + bd \end{aligned}$$

The product is therefore

$$P(a,b)P(c,d) = P(ac + bd,\ ad + bc + bd).$$

The answer is another PhiBase quantity.

Nothing has escaped from the arithmetic.

Nothing has been approximated.

The geometry quietly returned every occurrence of φ^2 to the same two building pieces with which we began.

One Example

Multiply

$$P(2,1)$$

by

$$P(3,2).$$

The calculation is simply a matter of counting.

$$\begin{aligned} \text{Short part} &= 2 \cdot 3 + 1 \cdot 2 = 8 \\ \text{Long part} &= 2 \cdot 2 + 1 \cdot 3 + 1 \cdot 2 = 9 \end{aligned}$$

Therefore

$$P(2,1)P(3,2) = P(8,9).$$

The same result appears if the two quantities are expanded in ordinary algebra. PhiBase has not changed the mathematics. It has preserved the construction.

Builder's Notebook

Multiply

$$P(1,1)P(1,1),$$

$$P(2,3)P(1,2),$$

and

$$P(4,1)P(2,2).$$

After each multiplication, identify the short part and the long part separately before writing the final PhiBase quantity.

Looking Ahead

Growth became simple because one quantity contained only a single nonzero part.

Can every multiplication be made to look that simple?

If we could find the right companion for a PhiBase quantity, one whole component of the product might disappear.

That question leads directly to division.

Stepping Stone 5

Undoing Multiplication

The previous stepping stone ended with a question.

Growing was easy because one part of the multiplication was already simple. One quantity contained only a single kind of piece, so the construction almost explained itself.

Can every multiplication be made to look like growth?

If the answer is yes, then division may not require a new idea at all. It may simply require finding the right companion.

Suppose we wish to multiply

$$P(a, b)$$

by another PhiBase quantity.

We are not interested in the whole product.

Not yet.

We care only about the long part.

From the multiplication rule,

$$P(a, b)P(x, y) = P(ax + by,\ ay + bx + by).$$

The second coordinate is the long part.

Everything becomes simple if that quantity can be made to disappear.

So the problem is no longer multiplication.

The problem is to choose

$$P(x, y)$$

so that

$$ay + bx + by = 0.$$

That is the bridge between multiplication and division.

Searching For A Playmate

Try a few examples.

Experiment.

Nothing prevents us from choosing different values for x and y and watching what happens.

Eventually one choice refuses to go away.

Let

$$x = a + b, \qquad y = -b.$$

Now substitute those values into the long part.

$$\begin{aligned} ay + bx + by &= a(-b) + b(a + b) + b(-b) \\ &= -ab + ab + b^2 - b^2 \\ &= 0. \end{aligned}$$

The long part has vanished.

The product now looks remarkably like the simple growth examples from the previous chapter.

We have found the playmate.

Mathematicians already have a name for it.

They call it the *conjugate*.

The conjugate of

$$P(a, b)$$

is

$$P(a + b, -b).$$

What Remains?

Once the long part has disappeared, only the short part is left.

Using the same substitution,

$$\begin{aligned} ax + by &= a(a + b) + b(-b) \\ &= a^2 + ab - b^2. \end{aligned}$$

Nothing mysterious remains.

The product has become

$$P(a, b)\, P(a + b, -b) = P(a^2 + ab - b^2, 0).$$

The remaining integer,

$$a^2 + ab - b^2,$$

is called the *norm*.

The conjugate removes the long part.

The norm is what is left behind.

Division

Now the original question answers itself.

To divide

$$P(u, v)$$

by

$$P(a, b),$$

multiply both numerator and denominator by the conjugate of the denominator.

The denominator immediately becomes the norm, an ordinary integer.

The numerator is an ordinary PhiBase multiplication.

Division has become the natural undoing of multiplication.

Nothing has been approximated.

Nothing has escaped from PhiBase.

Builder's Notebook

Find the conjugate of

$$P(2, 1).$$

Multiply the pair together.

Verify that the long part is zero.

Then divide

$$P(8, 9)$$

by

$$P(2, 1)$$

and confirm that the result is

$$P(3, 2).$$

Looking Ahead

The conjugate quietly introduces negative PhiBase coefficients.

The arithmetic has not changed.

The landscape has become larger.

The next stepping stone explores the path that has been hiding inside that landscape from the very beginning: the Fibonacci Spine.

Stepping Stone 6

The Fibonacci Spine

Division has completed the arithmetic.

Something else has quietly appeared along the way.

Every time we multiplied by φ, the same growth rule was applied. Every time we divided by φ, the same rule was reversed. We were no longer inventing new constructions. We were walking along a single path.

Begin with

$$P(1, 0) = 1.$$

One growth step gives

$$P(0, 1) = \varphi.$$

The next gives

$$P(1, 1) = \varphi^2.$$

Nothing changes except the counts.

Table 1 records the first few steps in both directions.

The Fibonacci numbers are easy to notice, but they are not the reason this table matters.

The table matters because every multiplication by φ has become one

Table 1: The Fibonacci Spine. Each row is obtained from its neighbour by one growth or one division step.

Power	PhiBase	Decimal
φ^{-4}	$P(5,-3)$	0.145898
φ^{-3}	$P(-3,2)$	0.236068
φ^{-2}	$P(2,-1)$	0.381966
φ^{-1}	$P(-1,1)$	0.618034
1	$P(1,0)$	1.000000
φ	$P(0,1)$	1.618034
φ^2	$P(1,1)$	2.618034
φ^3	$P(1,2)$	4.236068
φ^4	$P(2,3)$	6.854102
φ^5	$P(3,5)$	11.090170

step downward or upward through a prepared list. The arithmetic has become navigation.

Instead of multiplying by φ, move one row.

Instead of dividing by φ, move one row in the opposite direction.

The difficult work has already been done.

That is the beginning of efficient computation.

The Spine is not an approximation.

Every row is exact.

Every entry is a PhiBase quantity.

Every neighbour differs by exactly one growth step.

The arithmetic has become an index.

Builder's Notebook

Beginning with $P(1,0)$, generate six more rows using the growth rule.

Then begin again at $P(1,0)$ and walk downward by dividing by φ.

Compare your table with Table 1.

Looking Ahead

The Spine gives one exact path through PhiBase.

Most useful quantities do not lie on that path.

They lie between its rungs.

The next stepping stone builds the landscape around the Spine.

Stepping Stone 7

The Dense Grid

The Fibonacci Spine solved one important problem.

Repeated multiplication by φ no longer requires repeated arithmetic. It has become movement along an exact table.

That is wonderfully efficient.

Unfortunately, most PhiBase quantities are not powers of φ.

They live between the rungs of the Spine.

If the Spine is one exact path through PhiBase, what do we do with everything else?

We organize it.

Suppose we choose a coefficient bound B. Build every PhiBase quantity

$$P(n, p) = n + p\varphi$$

satisfying

$$|n| \leq B, \qquad |p| \leq B.$$

Keep only the positive values. Now sort them from the smallest to the largest.

Nothing has been approximated.

Nothing has been changed.

We have simply placed every exact quantity into its natural order.

That ordered collection is the *Dense Grid.*

Table 2: A small Dense Grid. Every entry is exact. The decimal value is shown only to illustrate the ordering.

Rank	PhiBase	Decimal
1	$P(5,-3)$	0.145898
2	$P(2,-1)$	0.381966
3	$P(-1,1)$	0.618034
4	$P(1,0)$	1.000000
5	$P(0,1)$	1.618034
6	$P(1,1)$	2.618034
7	$P(2,1)$	4.236068
8	$P(2,2)$	5.236068
9	$P(2,3)$	6.854102

The Spine passes through the Grid, but it does not define the Grid. Most entries are ordinary PhiBase quantities that simply happen to lie between neighbouring powers of φ.

The important idea is not the table itself.

It is that the ordering has already been done.

If two PhiBase quantities must be compared, the answer is already present.

If a nearby value is needed, its neighbours are already known.

The arithmetic has become memory.

The Spine remembers one exact family.

The Dense Grid remembers the landscape around it.

Together they remove a great deal of repeated work without sacrificing exactness.

Builder's Notebook

Choose

$$B = 2.$$

Construct every positive PhiBase quantity satisfying

$$|n| \leq 2, \qquad |p| \leq 2.$$

Sort them into ascending order.

Mark the entries that belong to the Fibonacci Spine.

Notice how the remaining entries fill the spaces between the Spine values.

Looking Ahead

The Spine and the Dense Grid now preserve everything we have learned.

The next question is no longer mathematical.

How can those two tables become a practical computational tool?

Stepping Stone 8

The PhiBase Slide Rule

The Fibonacci Spine changed something that we had almost taken for granted. Until that point, multiplying by φ meant performing another multiplication. After the Spine had been built, the same operation became nothing more than moving one row through a table. The arithmetic itself had not become simpler. We had simply remembered the result of every multiplication by φ before we needed it.

The Dense Grid quietly extends the same idea. The Spine remembers one remarkable family of PhiBase quantities, but most useful quantities are not exact powers of φ. They live between those familiar values. By arranging every exact PhiBase quantity into its natural order, the Dense Grid preserves the surrounding landscape just as faithfully as the Spine preserves its central path. Once that landscape has been explored, it never has to be explored again.

This is an old idea. Builders have always known that successful work should be remembered. A cabinet maker keeps patterns that have already proved themselves. A machinist keeps jigs because a carefully made tool should not have to be reinvented for every new part. An architect keeps drawings because a successful design is more valuable when it can be be used again tomorrow. Knowledge becomes useful not simply because it is correct, but because it remains available when the next decision must be made.

The PhiBase Slide Rule follows exactly that tradition. The Spine remembers every exact power of φ. The Dense Grid remembers the exact quantities that lie between those powers. Together they preserve relationships that would otherwise have to be reconstructed over and over again. The mathematics has not changed in the slightest. Only the place where the work is done has changed. The difficult thinking happens once. Every later calculation begins from that accumulated understanding.

That is why the Slide Rule is more than a table of numbers. It is a record of discoveries that no longer have to be rediscovered. Multiplication by φ becomes movement along the Spine. Comparison becomes movement through the Dense Grid. The arithmetic gradually changes from repeated computation into informed navigation, and that change opens the door to a different way of thinking about calculation.

The idea reaches beyond arithmetic. Whenever we preserve useful relationships instead of rebuilding them, tomorrow's work begins where yesterday's work ended. PhiBase simply carries that old engineering habit into exact arithmetic.

Builder's Notebook

Choose two neighbouring entries from the Fibonacci Spine and verify that moving one row upward multiplies by φ. Then choose two neighbouring entries from the Dense Grid and determine which is larger without performing any arithmetic. The answer has already been remembered for you.

Looking Ahead

The Slide Rule remembers exact arithmetic. The next question is whether the same philosophy can remember exact places. A point is more than a location. It is also the place from which the next step will begin. That is where PhiBase leaves arithmetic behind and begins to think like a builder.

Stepping Stone 9

The Flat Earth

The PhiBase Slide Rule preserves exact arithmetic by remembering relationships that have already been built. The next question is whether the same idea can preserve a place.

A length needs only one PhiBase quantity because every part of the construction lies on the same line. A flat surface is different. Once we step away from the line, we must remember not only how far we have travelled, but also the directions in which the travel occurred.

Two independent directions are enough to locate every point in a plane. Our turtle drawings have used this fact from the beginning. One basis vector points along 0°, and the other points along 72°. A point reached by the turtle is stored as an exact pair

$$\big(P(a, b),\ P(c, d)\big).$$

The first PhiBase quantity measures the contribution along the 0° direction. The second measures the contribution along the 72° direction. Every turn and every step is performed with exact PhiBase arithmetic. Floating point appears only when the finished construction is drawn on the page.

From the standpoint of representation, the problem is solved. Two independent basis vectors are sufficient.

Construction asks for something more.

Imagine a robot assembling golden triangles across a flat surface. While it works on one pentagonal region, one direction may be the most useful direction to follow. When it crosses into a neighbouring region, another direction may become the natural choice. A minimal two-vector description still locates the robot exactly, but it does not keep every useful construction direction ready for immediate use.

We therefore choose to remember five planar directions, separated by 72°.

The additional directions do not make the point more exact. They do not add another location. They preserve useful alternatives. When the preferred direction changes, the next direction is already part of the representation. The robot does not need to abandon its description and construct a new one before it can continue.

This is the difference between a minimal representation and a construction representation. A minimal representation keeps only what is necessary to describe the present point. A construction representation also keeps the directions that may become useful at the next boundary.

The five coordinates are therefore not five independent dimensions. The plane still has only two. The extra coordinates carry relationships among the five construction directions, allowing the same point to be understood from several useful orientations.

That choice gives the system a kind of nimbleness. The construction can change its local emphasis without losing the larger description. One direction may be central now, another a moment later, yet all five remain available within the same exact framework.

The Five Directions

Let the five planar construction directions be

$$\mathbf{E}_0,\ \mathbf{E}_1,\ \mathbf{E}_2,\ \mathbf{E}_3,\ \mathbf{E}_4,$$

with successive directions separated by 72°. A planar construction may then be recorded as

$$[q_0, q_1, q_2, q_3, q_4],$$

where each q_k is a PhiBase quantity associated with one construction direction.

Because the plane has only two independent dimensions, these five coordinates cannot vary independently. Their value lies in the simple transitions they preserve. A step that is awkward in one pair of directions may be natural in another, while the point itself remains unchanged.

The redundant description is not waste. It is stored opportunity.

Builder's Notebook

Begin with the two turtle basis directions,

$$0^\circ \qquad \text{and} \qquad 72^\circ.$$

Choose a point whose exact turtle coordinates are

$$\big(P(2,1),\ P(1,0)\big).$$

Now imagine that the construction crosses into a neighbouring pentagonal region. Which of the five directions would be most natural to follow next?

The point has not moved. Only the useful orientation has changed.

Looking Ahead

A plane needs only two independent directions, yet we remember five because construction benefits from having useful alternatives already present.

Space needs only three independent directions. For the same reason, the next stepping stone will remember six.

Stepping Stone 10

Building Space

When we were building on a line, life was simple. A single PhiBase quantity described every length because there was only one direction in which we could continue. The moment we stepped onto a flat surface, that simplicity disappeared. We could move in many directions, and our arithmetic had to learn how to remember places instead of lengths.

The ordinary mathematical answer was straightforward. Two independent directions are enough to describe every point on a plane, and our turtle drawings have quietly been using exactly that idea from the beginning. Every point in every figure is stored as two exact PhiBase quantities, one measured along the $0°$ direction and the other along the $72°$ direction. The mathematics remains exact until the final picture is drawn.

If our only goal were to describe where a point is, we could stop there.

Building asks a different question.

Imagine a robot laying golden triangles across a large floor. For a while one direction is the natural direction to follow. Then the robot reaches the edge of a pentagonal region and another direction immediately becomes the better choice. The point has not changed, but the construction has. The robot now has a different floor beneath its feet.

That small observation changes everything.

We are no longer asking how many directions are sufficient to describe a point.

We are asking how many useful directions should be remembered so that construction can continue naturally when the local floor changes.

Those are different questions.

Mathematics answers the first.

Construction answers the second.

That is why the previous stepping stone deliberately remembered five planar directions even though only two were mathematically necessary. The additional directions did not make the representation more accurate. They preserved useful possibilities. When the preferred direction changed, the next useful direction was already available. Nothing had to be rebuilt.

Space presents exactly the same choice.

Three independent directions are enough to describe every point in ordinary space. If our purpose were only representation, the familiar Cartesian coordinates

$$(x, y, z)$$

would already have solved the problem.

Our purpose is different.

We want a representation that grows naturally with the construction itself.

Instead of remembering only three directions, PhiBase remembers six construction floors. Each floor corresponds to one family of parallel dodecahedral faces. At any moment one floor is beneath the builder, while the other five remain immediately available should the construction cross onto another face.

A point is therefore described by six construction counts,

$$(a, b, c, d, e, f).$$

These are not six independent dimensions.

They are six remembered construction directions.

The point itself still occupies one place in ordinary three-dimensional space. The additional directions simply preserve the useful alternatives that make construction continuous instead of awkward.

The six construction floors are represented by the vectors

$$\begin{aligned}\mathbf{F}_1 &= (\varphi,\ 1,\ 0), & \mathbf{F}_2 &= (\varphi,\ -1,\ 0),\\ \mathbf{F}_3 &= (1,\ 0,\ \varphi), & \mathbf{F}_4 &= (-1,\ 0,\ \varphi),\\ \mathbf{F}_5 &= (0,\ \varphi,\ 1), & \mathbf{F}_6 &= (0,\ \varphi,\ -1).\end{aligned}$$

Beginning at the origin, the completed construction is

$$a\mathbf{F}_1 + b\mathbf{F}_2 + c\mathbf{F}_3 + d\mathbf{F}_4 + e\mathbf{F}_5 + f\mathbf{F}_6.$$

Collecting the components gives the ordinary Cartesian position,

$$\begin{aligned}x &= \varphi(a+b) + (c-d),\\ y &= (a-b) + \varphi(e+f),\\ z &= \varphi(c+d) + (e-f).\end{aligned}$$

Nothing has been lost.

The Cartesian coordinates tell us where the point is.

The six-floor description remembers how the construction reached that point and which neighbouring floors are immediately available when the construction continues.

One description serves the surveyor.

The other serves the builder.

Both describe exactly the same point.

Builder's Notebook

Begin with the construction

$$(1, 0, 0, 0, 0, 0).$$

Convert it into Cartesian coordinates.

Now imagine that the next step carries the construction onto a neighbouring dodecahedral face.

Which of the six construction floors becomes the natural one to continue from?

The point has not changed.

Only the floor beneath the builder has.

Looking Ahead

The builder's description and the surveyor's description describe the same point in different ways.

The next stepping stone shows how to move from one description to the other and back again without losing a single piece of information.

Stepping Stone 11

Finding Our Way Home

The previous stepping stone taught us that a builder and a surveyor look at the same world from different points of view. The builder remembers how the structure was assembled. The surveyor remembers where the finished structure stands. Neither description is better than the other, because each answers a different question.

A builder naturally asks,

> *How did I get here?*

A surveyor asks,

> *Where am I?*

Here both questions have exact answers, and we never leave the language of the dodecahedron to give them.

A dodecahedron shows six directions to the world — one for every pair of opposite faces. Any point we care to speak of is completely described by six numbers: how far the point lies from each of the six faces. We write them as

$$[a,\ b,\ c,\ d,\ e,\ f].$$

Each number is an exact golden-ratio quantity — never rounded, never approximated. Together, the six face-distances name the point exactly and completely. This is the surveyor's record.

The builder's record is different. It is the sequence of moves that reached the point. Each move steps along one of the six face-directions, and each move changes the six face-distances by a fixed, exact amount — that direction's own signature. To build is simply to add signatures. The six numbers gather the entire history of the construction, exactly, with nothing lost.

Building by Adding

Begin at the center, where every face is equally far away:

$$[z,\ z,\ z,\ z,\ z,\ z].$$

A single step along the first face-direction adds that direction's signature to the six numbers. A second step adds the same signature again. The construction grows, and the six numbers grow with it, and at every moment the surveyor can read the point's exact place straight from them.

Nothing has been approximated.

Nothing has been forgotten.

Coming Home

The real question is whether we can always find our way home. Suppose we have wandered deep into a construction and hold nothing but the six face-distances. Have we lost the way back?

Not at all. The six numbers *are* the way back.

To undo a move, subtract its signature. To return all the way to the beginning, subtract the moves in reverse order. Nothing is approximated along the way, because the six faces have been watching the whole time. A dodecahedron never forgets where its faces are, and so a point described by its six faces is never truly lost.

The Reward of Remembering the Faces

There is one more reward for speaking in the language of the six faces.

Suppose we want the mirror image of a point. A mirror of the dodecahedron does not measure and does not compute. It simply relabels the faces — sending each face to another, and for some, reversing a direction. And because our point is nothing but its six face-distances, the mirror image of the point is nothing but those same six numbers, shuffled the way the mirror shuffles the faces.

Here is one mirror of the dodecahedron:

$$[a,\ b,\ c,\ d,\ e,\ f] \longrightarrow [c,\ f,\ a,\ d,\ e,\ b].$$

Look closely at what it does. The distance to the first face becomes the distance to the third. The distance to the second face becomes the distance to the sixth. The fourth and fifth faces are left where they are. The mirror has reached across the solid and traded faces that are not opposite each other — face one for face three, face two for face six.

That reaching-across is the whole point. A mere flip along an axis can only trade a face for the face directly across from it. A true mirror of the dodecahedron does more: it carries a face to one of its *neighbors*. This is the five-fold nature of the solid showing itself — something no flat axis-flip can imitate. And still, the mirror image is found without a single measurement. We only rearranged the six numbers.

There are fifteen such mirrors in all, and each is its own shuffle of the six faces. Nothing is measured. Nothing is approximated. The builder reads the mirror's shuffle and applies it.

One Face, and a Warning

One caution completes the picture, and it is a gift rather than a nuisance.

Reflecting across the flat plane of a single face — not one of the dodecahedron's fifteen mirror symmetries, but one face's own plane — is a different act. When we try it, the arithmetic will not come out even: the answer must be divided by the number $\varphi+2$, and $\varphi+2$ refuses to divide our whole numbers cleanly, leaving a stubborn five behind. That stubborn divisor is a signal, kin to the scale-mismatch signal we have met before: it tells us the mirror image of that point does not land on the construction lattice at all.

So the clean reflections — the ones that stay exact and stay on the lattice — are exactly the mirror symmetries that shuffle the six faces.

When a reflection would carry us off the lattice, the arithmetic says so immediately and exactly, with that $\varphi+2$ underneath. We are never left guessing.

Builder's Notebook

Take a point given by its six face-distances, say

$$[1,\ 0,\ 2,\ 0,\ 0,\ 3].$$

Apply the mirror

$$[a,\ b,\ c,\ d,\ e,\ f]\ \longrightarrow\ [c,\ f,\ a,\ d,\ e,\ b]$$

by hand. Reading the rule off carefully, you should reach

$$[2,\ 3,\ 1,\ 0,\ 0,\ 0].$$

Notice that you never once had to measure anything, and notice *where the numbers went*: the value at face one landed on face three, and the value at face two landed on face six. The mirror carried each face to a neighbor, not to its opposite. That is a real dodecahedral reflection, and it is nothing more than moving six numbers around.

Now try to reflect the same point across a single face's own plane, and watch that $\varphi+2$ appear underneath, refusing to divide out. The arithmetic is telling you something exact: that image lies off the lattice.

The difficult part was choosing to remember all six faces. Once that is done, the mirror is almost effortless — and the arithmetic even warns us when a mirror would carry us away from the lattice.

Looking Ahead

We now have a language that measures exactly, remembers exactly, finds its way home exactly, and turns a point into its mirror image with very little effort — all without ever leaving the six faces of the dodecahedron.

But one move still eludes us: the *flip* — a mirror across a face's own plane — is the one that left that stubborn $\varphi+2$ behind, off the lattice. Is the flip truly beyond us, or is that divisor a door rather than a wall?

The next stepping stone opens it.

Stepping Stone 12

Look Hard at the Wall

The wall deserves a closer look.

Arithmetic told us only that the divisor was φ+2. It never explained why that number appeared, or why this one reflection should ask for it when every other symmetry of the dodecahedron was content with whole numbers.

Geometry has been waiting patiently to answer.

Look at the picture. Every turn in the pentagon has two parts. One reaches across the bottom, and one climbs the side. The horizontal part never causes trouble. Double the cosine of 36° and you obtain φ. Double the cosine of 72° and you obtain φ-1. Both belong to the golden arithmetic we already know. Every step across the bottom stays comfortably at home.

The climb is different.

Double the sine of 72°, and a new number appears. It is not golden. Let us call it ψ. The smaller climb is no different. Double the sine of 36° and it becomes (φ-1)ψ. Every height in the pentagon carries this same unfamiliar companion, quietly hiding where we had never thought to look.

Now the wall begins to look less like an accident.

The number we were searching for has been standing in the picture all along.

$$\psi^2 = \varphi + 2.$$

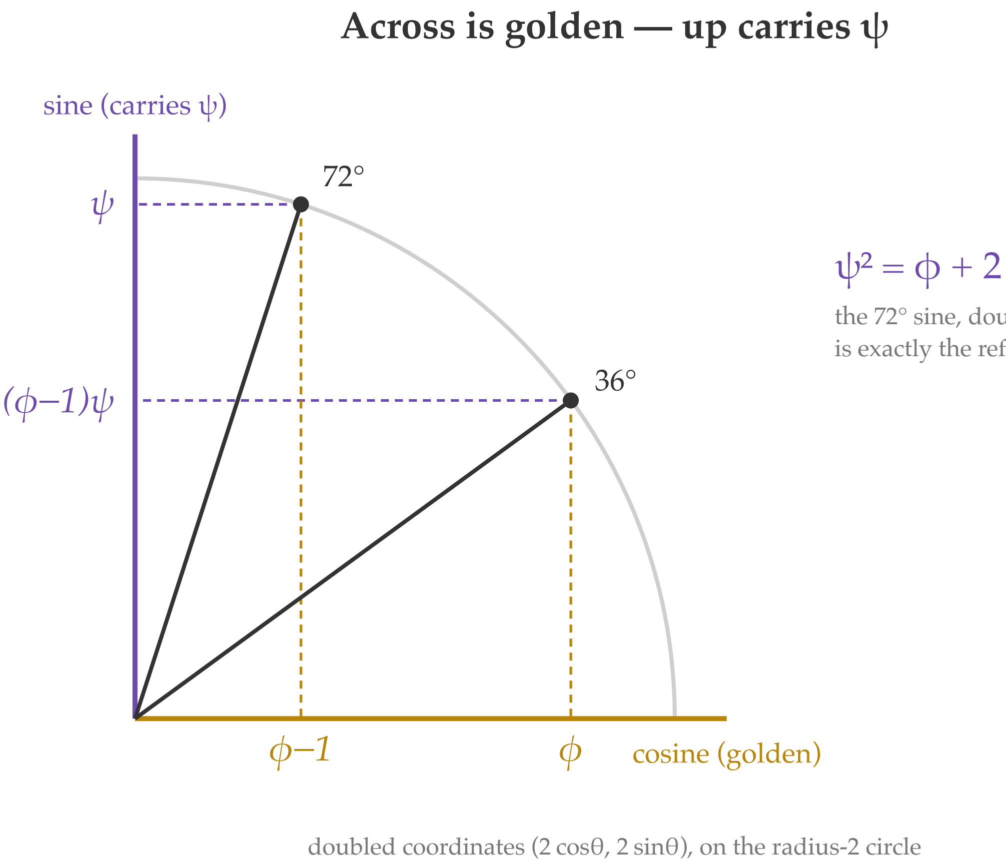

Figure 3: The 36° and 72° directions: cosines run golden across the bottom, sines climb the side carrying ψ.

The stubborn divisor is nothing more mysterious than the square of the pentagon's own height. The wall that stopped the flip was built by the pentagon itself.

Stepping Stone 13

Teaching a Machine

By the time we reach this point, the arithmetic is finished. We know how to measure, how to grow a construction, how to multiply and divide exactly, how to move through space, and how to reflect a structure without losing its exactness. The remaining question is no longer mathematical. It is practical.

Can a computer use this arithmetic just as faithfully as we do?

A computer does not understand mathematics. It does not recognize elegance, nor does it know whether one method is better than another. It simply performs the arithmetic it has been given, over and over again, without becoming tired or distracted. If the arithmetic is good, the computation is good. If the arithmetic is poor, the computer faithfully repeats the same poor method millions of times.

That observation shifts our attention away from the machine and back toward the arithmetic itself. Every computer program eventually becomes an enormous collection of very small operations. Numbers are added, subtracted, multiplied, divided, stored, and retrieved. The larger computation is nothing more than those small steps repeated many times.

Builders learned long ago that repeated work should be avoided whenever possible. A cabinet maker keeps templates. A machinist keeps jigs. An architect keeps drawings. The successful construction is preserved so that it never has to be rediscovered. Throughout this book the Fibonacci Spine, the Dense Grid, and the PhiBase Slide Rule have all followed that same idea. Do the difficult work once. Remember it. Reuse it whenever it is needed.

A computer is particularly good at remembering. Once an exact PhiBase relationship has been constructed, the machine can retrieve it whenever the same situation appears again. Nothing about the mathematics changes. Only the amount of repeated work changes.

This is not a new engineering idea. One of the most successful early microprocessors, the MOS 6502, contained no hardware multiplication instruction. Programmers frequently replaced repeated multiplication with carefully prepared lookup tables because memory was often less expensive than arithmetic. PhiBase follows exactly the same philosophy. The mathematics is performed once. The result is remembered. The computer simply repeats the construction.

Eventually every mathematical idea reaches the same moment. Someone builds it.

To answer the obvious question, an existing transformer inference engine was rebuilt so that its arithmetic used PhiBase instead of conventional floating-point multiplication. The model itself did not change. The training did not change. Only the arithmetic changed. The same inputs were presented, the same outputs were compared, and the resulting embeddings closely matched those of the original implementation while requiring substantially fewer arithmetic cycles.

That experiment does not prove that every future computer should use PhiBase. It answers a simpler question. Can a modern machine perform exact PhiBase arithmetic throughout a practical computation?

Yes.

Artificial intelligence is only one example. Robotic construction, structural analysis, crystal growth, and other problems built upon golden-ratio geometry are equally natural applications. Wherever the geometry belongs, the arithmetic can follow.

Builder's Notebook

Imagine two workshops. In the first, every successful measurement is repeated from the beginning each morning. In the second, every successful construction is carefully recorded and reused. Both workshops produce the same object. Which one would you rather work in? Now replace the workshop with a computer. Nothing else has changed.

Looking Ahead

We began with two simple triangles. We now have an arithmetic that both people and machines can follow exactly. One question remains.

What have we really been building?

Stepping Stone 14

The Builder's Way

When this journey began, we were not looking for a new arithmetic.

We were trying to understand why a handful of simple triangles could assemble themselves into structures of extraordinary beauty. The geometry kept asking questions that ordinary arithmetic could not answer. PhiBase was never the goal. It slowly became the answer.

That is not the path most mathematics books follow. They usually begin with definitions, introduce a collection of symbols, and then spend the rest of the book explaining what those symbols mean. We have travelled in the opposite direction. We built first. We watched carefully. Only after the ideas became familiar did we give them names.

Nothing in this book was introduced because mathematics happened to have a word for it. Addition appeared because builders join paths. Growth appeared because triangles become larger triangles. Multiplication appeared because builders combine lengths. Division appeared because every builder eventually needs to undo a construction. Even the conjugate was not invented to satisfy algebra. It appeared because the arithmetic itself demanded a companion that would make exact division possible.

Looking back, one simple habit guided every stepping stone. Whenever we built something, we immediately asked whether we could return home again. Addition was followed by subtraction. Growth by shrinking. Multiplication by division. The journey into space by the journey home. Reflection by reflection again. Reversibility became our measure of understanding. If a construction could not return home, we did not yet understand it well enough.

Along the way another idea quietly appeared. Every PhiBase number remembers how it was constructed. At first that seemed like nothing more than a convenient notation. Later it became an address. Then it became a point in space. Eventually it became an architectural language. The same idea never changed. It simply grew.

Perhaps that is the real lesson of this book.

Mathematics does not begin with symbols.

It begins with careful observation.

Someone notices a pattern.

Someone builds a simple example.

Someone asks one honest question after another.

The symbols come later.

They are not the discovery.

They are the memory of the discovery.

If you remember nothing else from these pages, remember this.

Do not be afraid to build.

Take things apart.

Put them together again.

Trust what survives repeated construction.

Give ideas names only after you understand them.

That is the Builder's Way.

One day you may discover a beautiful pattern that does not quite fit the mathematics you already know. When that happens, do not assume the mathematics is finished. Pick up a pencil. Build the simplest example you can. Watch it carefully. Let the construction teach you what the symbols have not yet learned to say.

That is how PhiBase began.

It is also how the next mathematics will begin.